

Steinitz exchange lemma

Lemma:

V : F -vector space

$Y, X \subseteq V$: finite subsets

s.t. (a) $\text{Span}(X) = V$

(b) Y is linearly independent

Suppose that Y has m elements

& X has n elements

Then:

(1) $m \leq n$

(2) If $Y = \{y_1, y_2, \dots, y_m\}$

then we have $X = \{\underbrace{x_1, x_2, \dots, x_m}_{\text{first } m \text{ elements}}, x_{m+1}, \dots, x_n\}$

s.t. the set $X' = \{y_1, y_2, \dots, y_m, x_{m+1}, \dots, x_n\}$
also spans V (i.e. $\text{span}(X') = V$)

P.f.: Proof will be by
induction on m

Base case ($m=0$) There is nothing
to show

Inductive hypothesis Y' : lin. ind. subset
 $\cap$
 $\cup$ with $m-1$

then $Y' = \{y_1, y_2, \dots, y_{m-1}\}$

$X = \{x_1, x_2, \dots, x_{m-1}, x_m, x_{m+1}, \dots, x_n\}$

s.t. $X'' = \{y_1, y_2, \dots, y_{m-1}, x_m, \dots, x_n\}$

satisfies

$$\text{Span}(X'') = V$$

Inductive step:

$$Y = \{y_1, y_2, \dots, y_{m-1}, \boxed{y_m}\}$$

$$Y' = \{y_1, y_2, \dots, y_{m-1}\}$$

$$X'' = \{y_1, y_2, \dots, y_{m-1}, \underbrace{x_m, \dots, x_n}_{\text{exchange with } y_m}\}$$

Need to know: We can exchange

one of these elements
with y_m without
changing spanning
property

We know:

(a) X'' spans V

i.e. Every $v \in V$ is a linear combination of

$$y_1, \dots, y_{m-1}, x_m, \dots, x_n$$

In particular

$$y_m = c_1 y_1 + c_2 y_2 + \dots + c_{m-1} y_{m-1}$$

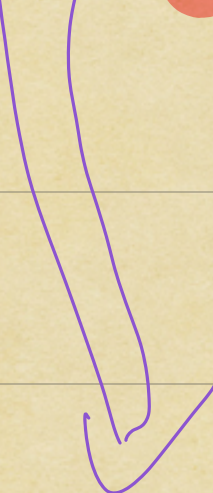
$$+ c_m x_m + \dots + c_n x_n$$

for $c_1, \dots, c_n \in F$

(b) Y is linearly independent

In particular,

y_m is not a linear combination of $y_1, y_2, \dots, y_{m-1}$


$$y_m = C_1 y_1 + C_2 y_2 + \dots + C_{m-1} y_{m-1} + C_m x_m + \dots + C_n x_n$$

for $C_1, \dots, C_n \in F$

and at least one of

$C_m, C_{m+1}, \dots, C_n$ is non-zero

Without loss of generality
(WLOG)

can assume $C_m \neq 0$

$\Downarrow C_m^{-1}$

$$C_m^{-1} y_m = C_m^{-1} c_1 y_1 + C_m^{-1} c_2 y_2 + \dots + C_m^{-1} c_{m-1} y_{m-1} \\ + x_m + \dots + C_m^{-1} c_n x_n$$



$$x_m = -C_m^{-1} c_1 y_1 - C_m^{-1} c_2 y_2 - \dots - C_m^{-1} c_{m-1} y_{m-1} \\ + C_m^{-1} y_m \\ - C_m^{-1} c_{m+1} x_{m+1} - \dots - C_m^{-1} c_n x_n$$

$\Rightarrow$ x_m is a linear combination
 of $y_1, \dots, y_m, x_{m+1}, \dots, x_n$

This tells me x''

$$\text{Span}(\{y_1, \dots, y_{m-1}, x_m, x_{m+1}, \dots, x_n\})$$

$$\subseteq \text{Span}(\{y_1, \dots, y_{m-1}, y_m, x_{m+1}, \dots, x_n\})$$

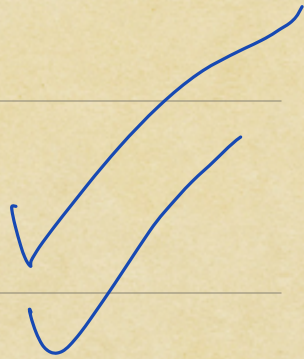
x'

$$\Rightarrow V = \text{Span}(x'')$$

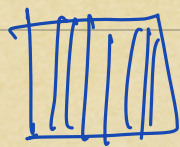
$$\subseteq \text{Span}(x')$$

$$\subseteq V$$

$$\Rightarrow \text{Span}(x') = V$$



Q.E.D



Defn V is a finite dimensional F -vector space if the following equivalent conditions hold:

(a) $\exists$ a finite subset

$$X \subseteq V \text{ s.t.}$$

$$\text{Span}(X) = V$$

(b) V admits a finite basis

Rmk (i) Equivalence follows from Corollary (i) from previous lecture

Prop (2) In this case, every basis for V has the same number of elements

Defⁿ V : finite dimensional
f.d.

The dimension of V ,
denoted $\dim(V)$,
is the size of any basis
for V .

Example:

$$(1) \dim(F^n) = n$$

$$(2) \dim(\text{Poly}_{\leq m}(F)) = m+1$$

A basis is given by

$$\{1, x, x^2, \dots, x^m\}$$

$m+1$ elements

$$(3) \dim(M_{r \times s}(F)) = r \cdot s$$

A basis is given by

$$\{E_{ij} : \begin{matrix} 1 \leq i \leq r \\ 1 \leq j \leq s \end{matrix}\}$$

The matrix with 1 in the i -th row

Δ j -th
Column

and 0s everywhere else

$$\underline{r=3, s=2}$$

$$E_{1,2} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Observation (1)

V : f.d. F -vector space

$W \subseteq V$: subspace

W is also finite dimensional

$$\& \dim(W) \leq \dim(V)$$

Observation (2)

In fact, if $\{w_1, w_2, \dots, w_m\}$ is a basis for W then, we can extend this to a basis

$\{w_1, w_2, \dots, w_m, v_{m+1}, \dots, v_n\}$

for V

Pf. Consequence of the
exchange lemma

Observation (3)

$$\dim(V) = n$$

$$\{v_1, v_2, \dots, v_n\} \subseteq V$$

TFAE : (1) $\{v_1, v_2, \dots, v_n\}$
is a basis

(2) $\{v_1, v_2, \dots, v_n\}$
is linearly
independent

(3) $\{v_1, v_2, \dots, v_n\}$
spans V

Pf. WTS: (2) $\iff$ (3)

"is equivalent to"

We know: There exists
a basis

$$\{w_1, w_2, \dots, w_n\} \text{ for } V$$

(This is the meaning of $\dim(V) = n$)

$$(2) \Rightarrow (3)$$

If $\{v_1, v_2, \dots, v_n\}$ is linearly independent

then the Steinitz Exchange Lemma
applied with $Y = \{v_1, v_2, \dots, v_n\}$

$$X = \{w_1, w_2, \dots, w_n\}$$

tells us $X' = \{v_1, v_2, \dots, v_n\}$ (all w_i 's exchanged!!)
 $= Y$ (!!)

satisfies $\text{span}(Y) = \text{span}(X') = \text{span}(X) = V$

$$(3) \Rightarrow (2)$$

If $\text{span}(\{v_1, v_2, \dots, v_n\}) = V$

We claim $\{v_1, v_2, \dots, v_n\}$ is linearly independent

Otherwise, WLOG, v_n is a linear combination of $v_1, v_2, \dots, v_{n-1}$

But then

$$\text{Span}(\{v_1, v_2, \dots, v_{n-1}\}) = \text{Span}(\{v_1, v_2, \dots, v_n\})$$

(why is this?)

This now tells us that V can be spanned by the $n-1$ elements $v_1, v_2, \dots, v_{n-1}$

But now applying Steinitz Exchange Lemma

to $X = \{v_1, v_2, \dots, v_{n-1}\}$

$Y = \{w_1, w_2, \dots, w_n\}$

we get the absurdity

$$n \leq n-1$$

XX

IMPOSSIBLE!

Defn V : f.d. F -vector space

$$B_1 = \{v_1, v_2, \dots, v_n\}$$

$$B_2 = \{w_1, w_2, \dots, w_n\}$$

are two bases for V

The change of basis matrix
from B_1 to B_2 is

the matrix $A \in M_{n \times n}(F)$

determined as follows:

$$A = (a_{i,j})_{\substack{1 \leq i \leq n \\ 1 \leq j \leq n}}$$

where $a_{i,j}$ are determined by.

$$W_i = a_{i,1} V_1 + a_{i,2} V_2 + \dots + a_{i,n} V_n$$

$$1 \leq i \leq n$$